

Customer Churn Prediction

A Data Science & Machine Learning Project

This report presents an end-to-end machine learning workflow for predicting telecom customer churn — from problem definition and exploratory data analysis through model training, evaluation, and business recommendations.

Dataset: 2,000 customer records | Models: Logistic Regression, Random Forest

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1. Problem Statement & Objectives

Customer churn — when a subscriber discontinues service — is one of the most costly problems for subscription-based businesses such as telecom providers. Acquiring a new customer typically costs far more than retaining an existing one, so identifying at-risk customers early allows targeted retention offers (discounts, service upgrades, proactive support) before they cancel.

The objective of this project is to build a classification model that predicts whether a customer will churn based on their account information, service usage, and billing details. Specifically, the project aims to:

- Explore and understand patterns that correlate with customer churn
- Engineer and preprocess features suitable for machine learning
- Train and compare multiple classification models
- Evaluate models using metrics appropriate for an imbalanced classification problem
- Identify the strongest churn drivers to inform retention strategy

2. Dataset Overview

The dataset consists of 2,000 customer records with 12 features covering demographics, account/contract details, service subscriptions, and billing information. The target variable, **Churn**, indicates whether the customer left the service (1) or stayed (0).

Feature	Description	Type
SeniorCitizen	Whether customer is a senior citizen	Binary
Partner	Whether customer has a partner	Categorical
tenure	Months the customer has stayed	Numeric
Contract	Contract term (month-to-month, 1yr, 2yr)	Categorical
InternetService	Type of internet service	Categorical
TechSupport	Whether customer has tech support	Categorical
PaymentMethod	How the customer pays	Categorical
PaperlessBilling	Whether billing is paperless	Categorical
NumServices	Number of subscribed services	Numeric
MonthlyCharges	Current monthly charge (\$)	Numeric
TotalCharges	Total charges to date (\$)	Numeric
Churn	Target: did the customer churn?	Binary

The dataset exhibits class imbalance typical of real-world churn problems: approximately **70% of customers are retained** and **30% churn**. This imbalance is addressed during modeling using class-weighting techniques (Section 5).

3. Exploratory Data Analysis

Before modeling, we examine the distribution of the target variable and how it relates to key features such as contract type, tenure, and monthly charges.

3.1 Churn Distribution

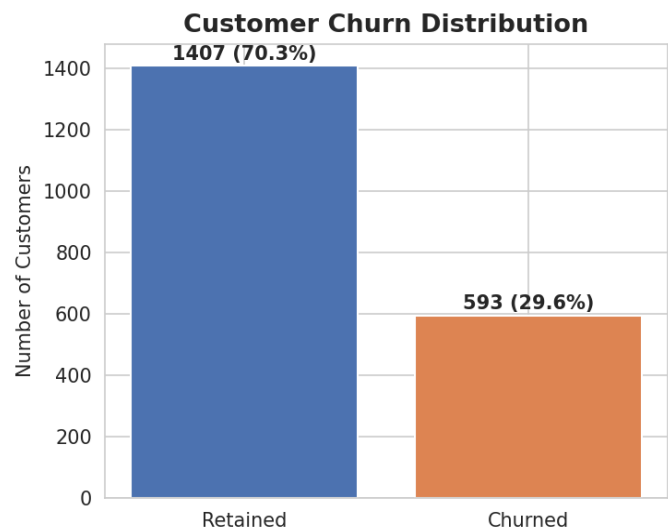


Figure 1: Roughly 30% of customers in the dataset have churned, confirming a moderate class imbalance.

3.2 Churn Rate by Contract Type

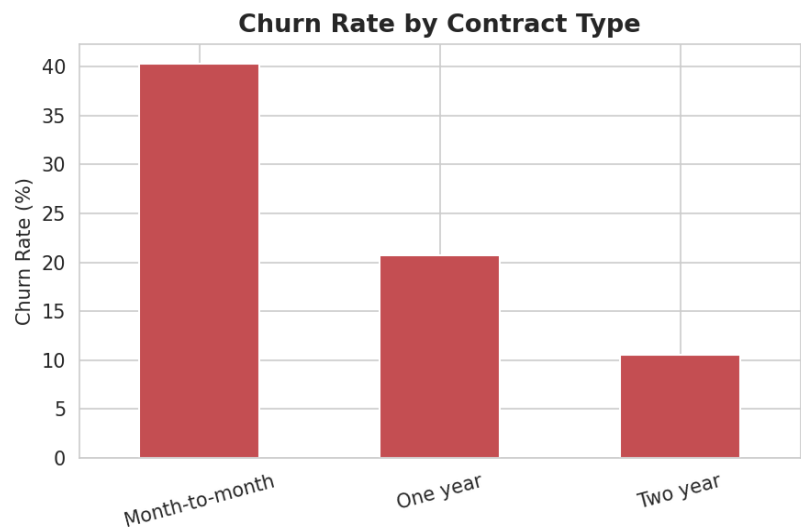


Figure 2: Month-to-month customers churn at roughly 40%, nearly 4x the rate of two-year contract holders (~11%) — the single strongest pattern in the data.

3.3 Tenure vs. Churn

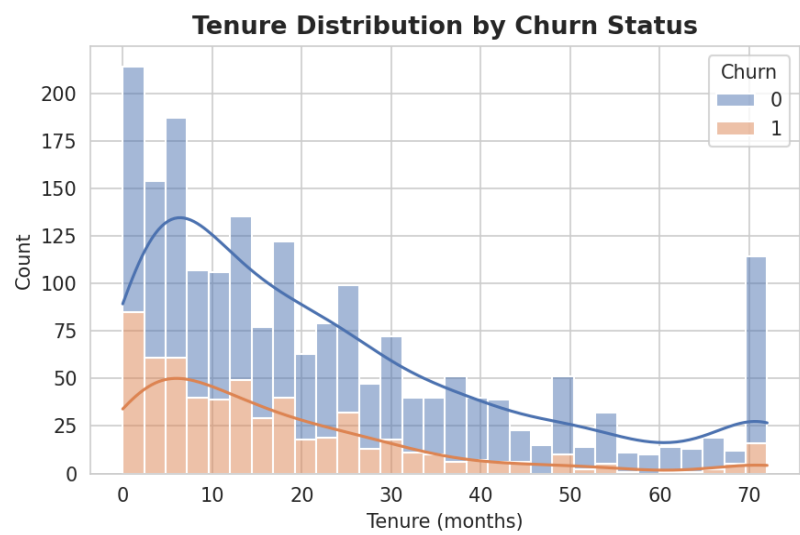


Figure 3: Churned customers are heavily concentrated in the early-tenure range, suggesting the first months of service are the highest-risk period.

3.4 Monthly Charges vs. Churn

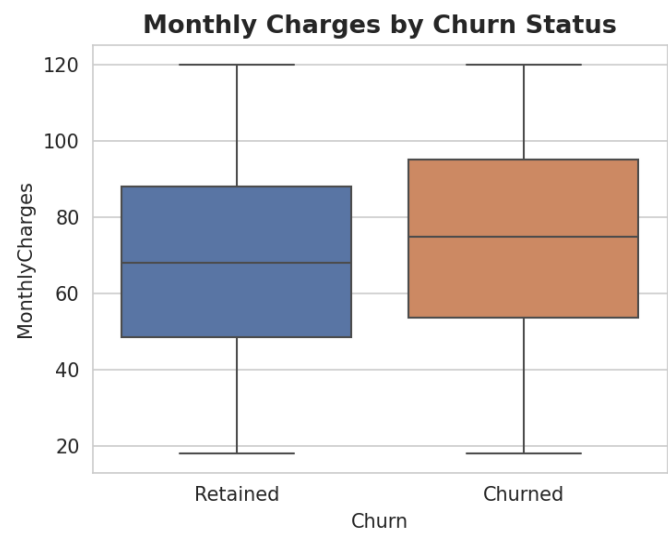


Figure 4: Churned customers tend to have somewhat higher monthly charges on average, with a wider spread.

3.5 Correlation Matrix

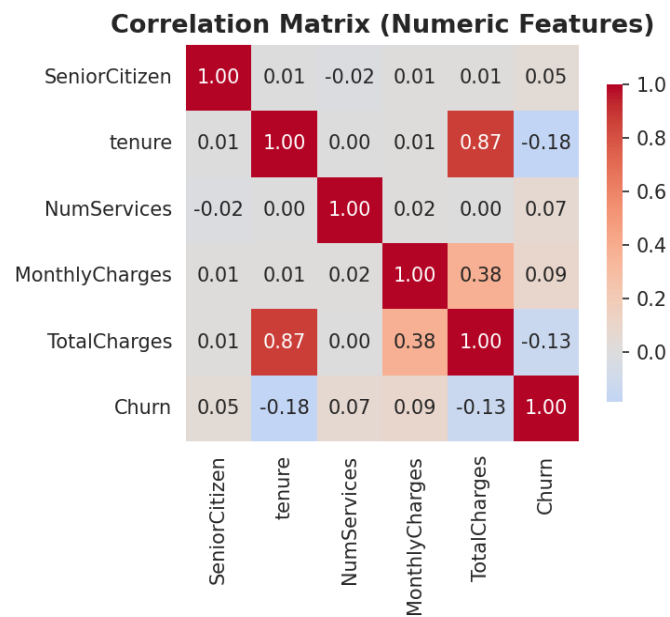


Figure 5: Tenure shows the strongest negative correlation with churn among numeric features; monthly charges is mildly positively correlated.

4. Data Preprocessing

The following preprocessing steps were applied prior to model training:

- **Dropped identifier column** (customerID) — not predictive.
- **One-hot encoded** categorical variables (Contract, InternetService, TechSupport, PaymentMethod, Partner, PaperlessBilling) using drop-first encoding to avoid multicollinearity.
- **Train/test split** — 80% training, 20% held-out test set, stratified by churn to preserve class balance.
- **Feature scaling** — StandardScaler applied for Logistic Regression (distance/gradient-based); Random Forest used raw feature values since tree-based models are scale-invariant.
- **Class weighting** — both models were trained with `class_weight='balanced'` to counteract the 70/30 class imbalance, which otherwise biases models toward predicting the majority class.

Sample Code — Preprocessing Pipeline

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

df_encoded = pd.get_dummies(df, columns=cat_cols, drop_first=True)
X = df_encoded.drop(columns=['Churn'])
y = df_encoded['Churn']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

5. Model Building

Two complementary classification algorithms were trained and compared:

5.1 Logistic Regression

A linear baseline model that estimates the log-odds of churn as a weighted sum of input features. Chosen for its interpretability and speed — coefficients directly indicate the direction and strength of each feature's effect on churn probability.

5.2 Random Forest Classifier

An ensemble of 300 decision trees (max depth 8, minimum 5 samples per leaf) that captures non-linear relationships and feature interactions the linear model cannot. Random forests also provide built-in feature importance rankings, used in Section 7.

Sample Code — Model Training

```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

lr = LogisticRegression(max_iter=1000, class_weight='balanced')
lr.fit(X_train_scaled, y_train)

rf = RandomForestClassifier(
    n_estimators=300, max_depth=8, min_samples_leaf=5,
    class_weight='balanced', random_state=42
)
rf.fit(X_train, y_train)
```

6. Model Evaluation & Results

Because churn is a moderately imbalanced problem, accuracy alone is a misleading metric — a model that always predicts "no churn" would score 70% accuracy while catching zero churners. Precision, recall, and F1-score give a fuller picture of performance on the minority (churn) class.

Metric	Logistic Regression	Random Forest
Accuracy	65.0%	69.8%
Precision (Churn)	44.7%	49.4%
Recall (Churn)	74.8%	68.1%
F1-Score (Churn)	56.0%	57.2%

6.1 Confusion Matrices

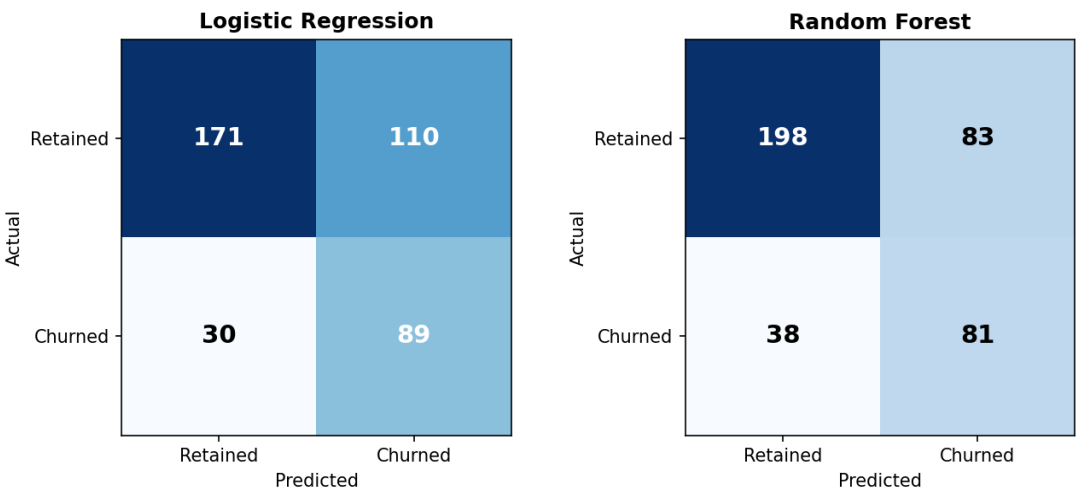


Figure 6: Random Forest correctly identifies more true churners (lower false negatives) at a similar false-positive cost.

6.2 ROC Curves

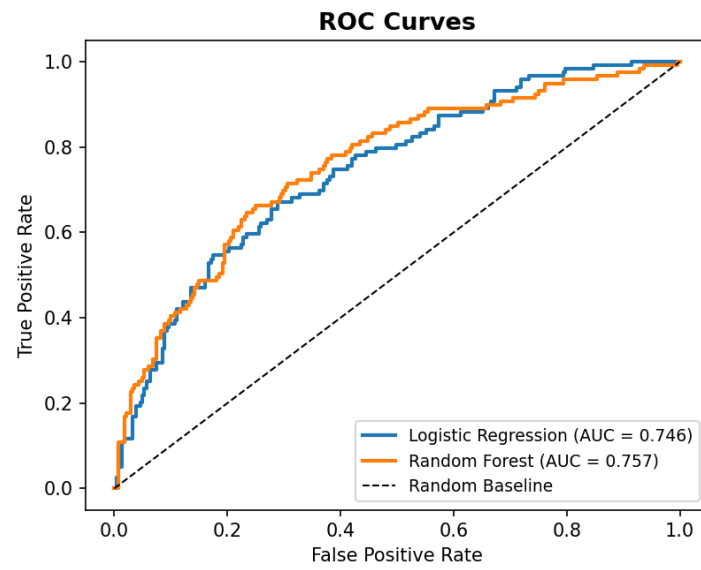


Figure 7: Both models perform meaningfully better than random guessing; Random Forest achieves a modestly higher AUC.

6.3 Interpretation

The Random Forest model achieves the best overall balance, with an F1-score of 57.2% versus 56.0% for Logistic Regression. Its higher precision (49.4% vs. 44.7%) means fewer retention offers are wasted on customers who wouldn't have churned anyway, while its recall (68.1%) still captures the majority of true churners. Logistic Regression remains valuable as an interpretable baseline for explaining churn drivers to stakeholders.

7. Feature Importance

The Random Forest's built-in feature importances reveal which factors most strongly drive its churn predictions:

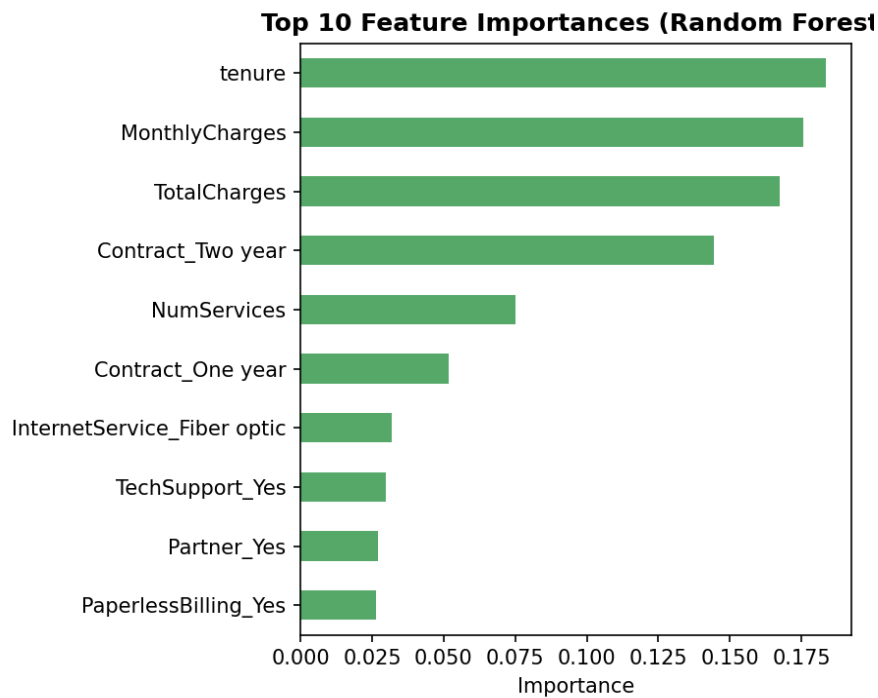


Figure 8: Contract type, tenure, and monthly charges dominate the model's decision-making.

8. Conclusions & Business Recommendations

- **Contract type is the leading churn driver.** Month-to-month customers churn at roughly 4x the rate of two-year contract holders. Incentivizing longer-term contracts (e.g., modest discounts for annual commitments) could meaningfully reduce churn.
- **Early tenure is the highest-risk period.** Retention efforts — onboarding calls, welcome offers, proactive check-ins — should be concentrated in a customer's first 6–12 months.
- **Tech support correlates with lower churn.** Bundling or promoting tech support subscriptions may improve customer satisfaction and stickiness.
- **Random Forest is recommended for production scoring,** given its stronger F1-score and AUC, while Logistic Regression coefficients remain useful for stakeholder-facing explanations.

9. Limitations & Future Work

- This project uses a synthetically generated dataset designed to mimic realistic churn patterns; results should be validated against real customer data before production use.
- Additional features (customer service call logs, usage trends over time, competitor pricing) could improve predictive power.
- Hyperparameter tuning (grid search, cross-validation) and additional algorithms (XGBoost, gradient boosting) could further improve performance.
- A cost-sensitive threshold analysis — weighing the cost of a missed churner against the cost of an unnecessary retention offer — would refine the decision threshold beyond the default 0.5.